# Calculations Related To The Microscope (Part 2)

Unit 1: Introduction to Biology
Module 2: The Microscope
Learning Objective(s): 2.4, 2.5
Lesson 4: Estimating the Size of an Object (Calculations – Part 2) and Preparing Biological Drawings
Lesson Objective(s):
- To become familiar with the procedure used for estimating the size of objects observed.
- To become familiar with the methods and components of a biological drawing.

Evaluation: Assignment: [15 marks]
- Microscope Calculation – Worksheet One [5 marks]
- Microscope Calculation – Worksheet Two [10 marks]

Once you have completed the calculations used in Lessons 2 and 3 (Microscope Calculations #1-3), it is possible to estimate the size of any object that you are viewing on the microscope that you used for the above calculations.

Microscope Calculations (continued)

4) Estimating the size of an object:
- You must first calculate the diameter of the field of view for the power of magnification you are using.
- Looking through the eyepiece, estimate how many times the object will fit across the field of view.
- Calculate the size of the object using the formula below.
- Remember to use um

Example:
The image of the amoeba to the left is seen under medium power (100X magnification).
If the diameter of the field of view on low power (40X) for this microscope is 4000 µm, first calculate the diameter of the field of view for medium power as follows:

Second, estimate the number of times the object (amoeba) fits across the field of view. Unless otherwise instructed, estimate using the longest dimension of the object. In the image to the left, the amoeba would fit across the field of view approximately 2 times. Because the diameter of the field of view for medium power was calculated as 1600 µm, the size of the object would be calculated as follows:

Therefore, the approximate length of the amoeba is 800 µm or 0.8mm.

The final calculation you will need to use when using the microscope is to calculate the magnification of your drawing. (This is a way of calculating how many times larger your drawing is in relation to the estimated size of the object.)
Before calculating the magnification of a drawing, you need to review the proper methods and components used in a biological drawing. See "Biological Drawings" for further instruction.

5) Calculating the magnification of your drawing:
- First calculate the size of the object using the procedure outlined above.
- Using a ruler, measure the size of your drawing and convert units to um. (In the diagram below a ruler has been superimposed on the drawing to

illustrate how to measure. The image in this particular example is 13.7cm or 137mm, which converts to 137 000 µm.)
Calculate the magnification of your drawing using the formula below the diagram.

This number indicates how many times larger your drawing is relative to the actual size of the object.
This number should appear at the bottom of your drawing.
Now put all of your skills together and complete the following worksheets, which are to be submitted for evaluation.

Microscope Calculations – Worksheet one

Microscope Calculations – Worksheet two

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